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Helping Teachers Harmonize: Supporting Teachers in Creating Customized, Localized, Culturally Relevant Instructional Materials

DAVID WEINTROP
University of Maryland, USA
weintrop@umd.edu

DIANA FRANKLIN
University of Chicago, USA
dmfranklin@uchicago.edu

MINH TRAN
University of Chicago, USA
ngminhtran@uchicago.edu

JENNIFER PALMER
University of Chicago, USA
jenpalmer@uchicago.edu

Customizing instructional materials to draw on the prior knowledge and lived experiences of students is an important but time-intensive process. To help accomplish the goal of creating personalized, localized instructional materials in an efficient and manageable way, we introduce the harmonizing process. Harmonizing is a structured, scaffolded process that helps teachers revise existing instructional materials to better resonate with the identities, interests, and ideas of students while preserving the focal learning goals of the lesson. This paper introduces the motivation and big ideas of the harmonizing process, along with the strategies, technological tools, and scaffolds developed for carrying it out. To provide a concrete demonstration of the harmonizing process in action, we share examples of teacher-harmonized middle

school computer science materials. In doing so, it provides a useful design process to be added to the tool belt of teachers and curriculum designers to help align instructional materials with the students who will learn with them.

INTRODUCTION

Situating learning opportunities in contexts that are familiar to learners and that draw on their existing knowledge and prior experience can help them learn the desired content and increase their sense of self-efficacy (Gay, 2010; Ladson-Billings, 2014; Moll et al., 1992; National Research Council, 2018). Customizing instructional materials to accomplish this has long been a goal of educators and instructional designers. Identifying familiar and compelling contexts in which to situate instruction is a critical task for teachers, particularly in disciplines that can be seen as "decontextualized"—where the content might not directly connect to learners' lived experiences. For example, research in mathematics education has shown how students perform better on relevant and contextualized questions (e.g., "If a basketball player makes 6 out of 10 free throws, what is their free throw percentage?") compared to isomorphic, decontextualized versions of the same question (e.g., 6/10 = ___%") (Nasir & Hand, 2008).

In response to these findings, curriculum designers and educators have developed curricula and instructional materials designed to resonate with learners' interests, experiences, and prior knowledge. This work has a long history across numerous disciplines, including the foundational work of Moll and colleagues on Funds of Knowledge in mathematics (1992) and Lee (1995) and colleagues in English language arts. Those efforts, and others like them, were created with specific communities in mind and demonstrate the impact of designing with and for the target audiences.

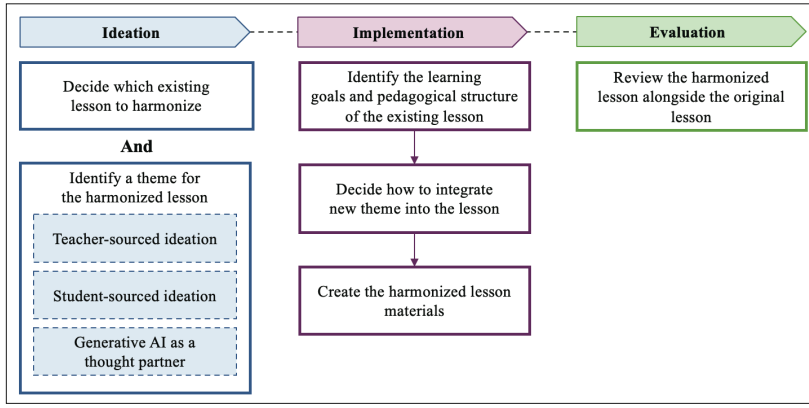
Given the importance of contextualizing instructional materials within students' prior knowledge and experiences, what can an educator do if no existing curriculum fits their instructional needs or student population? One solution is to develop ways to support teachers in doing this curricular customization work. As experts in their classrooms, teachers know their students, schools, and communities best. In this work, we present a design process intended to support teachers in using their knowledge to contextualize instructional materials for their students. The motivating question for this work is: How can we support teachers in creating customized, local-

ized, culturally responsive instructional materials that draw on the prior knowledge and cultural resources present in their classrooms?

This paper presents our answer to this question: an instructional design process we call harmonizing. The term harmonizing captures the act of starting with an existing artifact (e.g., a song or an instructional activity) and then creating a complementary version that resonates with the original, not changing its core purpose or structure, but making it unique and personalized. In this work, we present a harmonizing process educators can employ to customize instructional materials to resonate with lived experiences and draw on prior knowledge of their students. The harmonizing process consists of three steps: ideation, implementation, and evaluation (Figure 1). For each step, we have developed scaffolds and strategies to assist educators and instructional designers in carrying out the harmonizing process. For the ideation step in particular, we have identified three distinct strategies to help teachers harmonize classroom materials. The first strategy supports teacher-generated ideas, the second supports student-generated ideas, and the third uses generative AI as a thought partner in developing ideas and classroom resources.

Figure 1

Graphical Representation of the Harmonizing Process



The central goals of this manuscript are to (a) introduce the idea of harmonizing existing materials; (b) define the three steps of the harmonization process; (c) present three distinct harmonizing ideation strategies and their accompanying scaffolds; and (d) provide examples of teacher-created

harmonized materials, using a middle school computer science curriculum as the source curriculum. In doing so, this work contributes a harmonizing process to teachers' and teacher educators' tool belts, including a set of strategies and scaffolding resources to empower teachers to create customized and localized instructional materials for their students. This approach leverages the expertise of teachers, prior knowledge of students, and cultural resources present in classrooms to create effective, accessible, and equitable learning experiences. Further, this work presents a set of scaffolds to help teachers carry out the harmonizing process. Collectively, this work seeks to equip teachers and teacher educators with strategies and tools to help create classroom instructional materials that draw on the prior knowledge and localized cultural resources of the classrooms in which they are taught.

This paper continues with a review of prior work that informs the harmonizing process. Next, we briefly present the curriculum and professional development context that will be used to provide illustrative examples of harmonized materials throughout the remainder of the paper. We then present the harmonizing process, including the three distinct ideation strategies and the design scaffolds that accompany each. The paper concludes with a discussion of the contributions and implications of this work.

PRIOR WORK

Resource Pedagogies and Tailoring Instructional Materials

Drawing on the cultural knowledge and lived experiences of students can positively impact their learning (National Research Council, 2018). Captured under the umbrella term resource pedagogies, various frameworks and curricula have been designed to support the academic success of students from racially, ethnically, and linguistically marginalized communities, promoting equity in teaching practices (Lee, 2007; Moll et al., 1992; Nasir et al., 2006). Among others, culturally relevant pedagogy (Ladson-Billings, 2014), culturally sustaining pedagogy (Paris, 2012), and culturally responsive computing (Scott et al., 2015; Scott & White, 2013) aim to leverage students' existing knowledge, including their languages, cultures, lived experiences, and identities as assets that are essential foundations for learning (Castro, 2022; Gay, 2002; Ladson-Billings, 2014).

Resource pedagogies encourage a tailored approach to teaching and learning, where curriculum designers, educational researchers, or educa-

tors customize content to resonate with the lived realities, identities, and cultural and linguistic repertoires of their students (Madkins et al., 2023; Reyes & Norman, 2021). Teachers may engage in relationship-building with students to identify what students care about and what assets they bring to their learning. Then, they can use these assets and relationships to design culturally relevant learning experiences. Additionally, resource pedagogies encourage the deployment of customizable instructional practices while being flexible in format and outcome. Doing so requires recognizing the role of identity in students' learning, and centering identity in instructional materials and approaches (Barab & Luehmann, 2003; Gerard et al., 2010). Resource pedagogies are also an opportunity to resist deficit narratives and amplify asset-based narratives that elevate student voices and agency, value community practices to inspire curriculum, and seek justice and representation through design (Alvarez, 2021; Cruz-Gonzalez, 2024).

Connecting instructional materials to learners' cultures, prior knowledge, and lived experiences is critically important for developing positive and productive learning spaces for students (National Research Council, 2018). Building these connections can help to motivate learners, connect content to their personal goals and interests, and develop confidence and a sense of belonging in academic spaces. Aligning formal instructional goals with the values of a student and their communities can help motivate them to meaningfully engage in the learning process (Eccles et al., 1983; Wigfield & Eccles, 2000). Connecting with learners' interests can serve as a powerful motivator for learning (Michaelis & Weintrop, 2022; Renninger & Hidi, 2015). Further, building these connections can help deepen learners' commitment to the learning process, build confidence and self-efficacy, and increase expectations of being able to succeed in learning and using the materials being taught (Bandura, 1977).

In computer science (CS) education, incorporating resource pedagogies includes customizing instructional materials by incorporating cultural practices through family stories, real-world issues, and languages that resonate with students, enabling them to develop critical consciousness about themselves and others (Matuk et al., 1998; Narayanan, 2024). For example, in their study of belonging and well-being with Latine youth in resource pedagogy, Kaneria et al. (2024) found that students experienced increased persistence in their CS identity and knowledge when situating instruction in the musical traditions of students' different cultures. Relatedly, Novak and Khan (2018) argue that CS curricula should leverage children's everyday experiences and make use of students' preferences for individual topics and learning activities. The tailoring approach positions CS as a tool for critical

inquiry, harnessing connections between students and coding, and allowing students to bring their interests to computing classrooms (Padró-Collazo et al., 2023). Collectively, this work shows how student-driven personalization of learning is an important mechanism for employing resource pedagogies and the potential they have in CS education.

Supporting Teachers in Customizing Instructional Materials

Teachers creating and integrating customized learning materials and resources can help them connect with their students and produce equitable educational opportunities for students (Almuhanna, 2024; Sumner, 2010). As Wang et al. (2021) noted, customizable instructional materials are most effective for teachers and students when they can be adapted to resonate with students' interests and increase engagement. Further, Gerard et al. (2022) found that customizing instructional materials can foster self-directed learning in students. For teachers, being supported in customizing existing instructional materials can be more helpful than having to create brand-new materials. Teacher-customized instructional materials greatly improve students' content area understanding as compared to non-customized materials, leading to an improved quality of inquiry-based lessons that center on students' questions and curiosity (Littenberg-Tobias et al., 2016; McNeill et al., 2024). Students actively explore concepts, ask questions, and seek answers through participation, observation, and experimentation. The customized materials explored in this work are derived from instructional materials designed to encourage students' critical thinking and center student-driven learning, with teachers acting as facilitators rather than directing students.

Customizable CS education centers learning experiences to meet the needs, backgrounds, interests, and contexts of diverse students. Instead of a one-size-fits-all curriculum, students engage with adaptable activities, projects, and tools that reflect their identities, communities, and goals. However, the process of customizing instructional materials can be time-consuming and challenging for teachers to complete while handling other tasks during the school year (Tran et al., 2024). One emerging strategy to facilitate this process is the use of generative AI. Generative AI tools can help improve the efficiency of teachers working through the customization process by using modeling and brainstorming strategies (Pesovski, et al. 2024; Tran et al., 2025). These tools can also be used in conjunction with students, inviting them to co-create instructional materials, which allows students to navi-

gate through the content in a way that reflects their unique learning needs and aspirations (Littenberg-Tobias et al., 2016; McNeill et al., 2024; Wang et al., 2021).

Culturally Responsive K–12 Computer Science Instruction

The last two decades have seen a rapid growth in interest in K–12 CS education globally (Anwar et al., 2020; Hubwieser et al., 2015; Tshukudu et al., 2023). This growth has prompted both educators and policymakers to redefine the goals of CS instruction and reconsider the most effective ways to teach computational thinking (Bocconi et al., 2016; Grover & Pea, 2013; Tissenbaum et al., 2021). The result has been both national CS curricula in countries such as New Zealand and England, as well as guidelines for CS in regions that adopt a more decentralized approach, like the United States and India (Bell et al., 2010; Crick & Sentance, 2011; Raman et al., 2015; Wang et al., 2016). Relatedly, CS teacher preparation efforts are underway to support the deployment of large-scale CS initiatives. However, the contents and goals of these efforts vary in core content knowledge, teacher preparation pathways, and attention to issues of diversity, equity, and inclusion by region (Yadav et al., 2022).

Broadening participation in the field has been a central goal of many CS education efforts (Leftwich et al., 2020), seeking to directly address longstanding issues of underrepresentation and exclusion of certain populations, including female, non-binary, BIPOC, non-native English speakers, and neurodivergent students (Childs et al., 2024). As part of this push, curriculum designers, standards authors, and advocacy groups have collectively been pushing to more directly attend to issues of access and equity as part of K–12 education (CSTA Standards Task Force, 2017; *K–12 Computer Science Framework*, 2016). Teachers play a critical role in carrying out this work as they must "proactively counter stereotypes that exclude students from opportunities to excel in computing" (ISTE, 2021, p. 11). In response, educators and researchers have begun to incorporate resource pedagogies into CS instruction (Leonard & Sentance, 2021).

Culturally responsive computing approaches focus on how teachers engage with students' cultural backgrounds to promote academic success, foster positive relationships, affirm and sustain cultural identities, and confront systems of inequality that disadvantage marginalized groups (Madkins et al., 2020). As part of this work, Madkins et al. describe three intersecting but distinct practices for equitable computing instruction: (a) supporting

students' identity development by recognizing who they are and encouraging self-expression through computing; (b) emphasizing the social and personal significance of technology by connecting it to local and sociopolitical contexts; and, (c) empowering students as innovators and change-makers who use technology to tackle personally relevant challenges. This work emphasizes the importance of students expressing their cultural identities in computing work, while also developing a strong sense of belonging within the field, and marks a shift from deficit-based approaches that frame students from marginalized groups as lacking essential skills and in need of remedial, decontextualized instruction to centering students' identities, cultures, and lived experiences as valuable assets in the computing classroom. This allows students to engage with technology in ways that are personally meaningful and culturally relevant, fostering a sense of connection between their communities and the discipline of computing (Leonard & Sentance, 2021; Scott et al., 2015). Having reviewed the relevant literature, we briefly present the curriculum and professional development context that we will use to assist in our presentation of the harmonizing process. After that, we present the harmonizing process itself. Along the way, we will present illustrative examples from the teachers who participated in our professional development session to demonstrate what each step/strategy in the harmonizing process looks like in action.

CURRICULUM AND PROFESSIONAL DEVELOPMENT CONTEXT

Before presenting the harmonizing process itself, we first briefly introduce the curriculum and teacher professional development context that we will use to provide illustrative examples of the harmonizing process throughout the remainder of the manuscript. The goal of doing so is to be able to provide concrete, teacher-created examples of the harmonizing process in action.

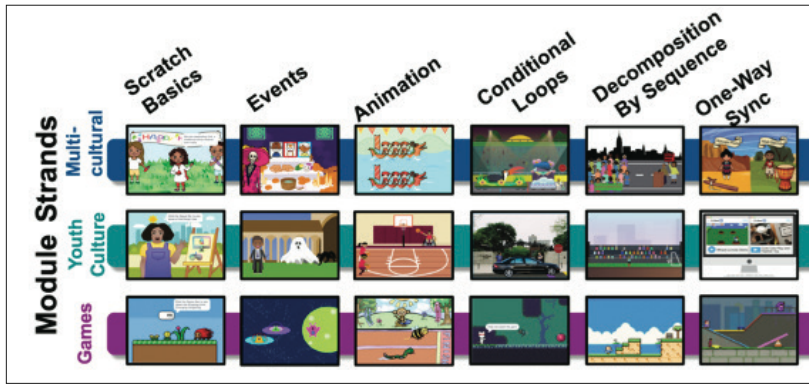
Meet Scratch Encore

Scratch Encore is a fully-featured intermediate middle-school CS curriculum (Franklin et al., 2020). It consists of 15 modules that constitute a computing learning trajectory, moving from basic computing concepts such as events and loops to more advanced topics like initialization and state.

The computing content is taught using Scratch (Resnick et al., 2009), a visual block-based programming tool, with the context of the content being conveyed through visual assets (e.g., sprites and backgrounds in the Scratch Environment) and project actions (e.g., sprite dialog and movement). Each module follows the same pedagogical structure, with the modules being organized across a set of thematic strands.

Figure 2

The Strands (Themed Rows) and Modules (Content-Based Columns) for the First Six Modules of Encore



Scratch Encore employs a strand/module structure (Figure 2). Each module comprises several lessons and covers a different computing topic (shown as columns in Figure 2). Each module can be taught using one of three strands (shown as rows in Figure 2): multicultural, youth culture, and games. The multicultural strand draws on celebrations, events, or traditions, especially those of minority cultures. The youth culture strand includes themes that resonate with elementary and middle school-aged learners, including sports and social networks. The games strand builds game elements, resulting in playful, engaging Scratch programs. Each strand situates the content of the activities within a different context while covering the same CS material. For example, students can learn about events in Scratch by creating an Ofrenda as part of a Día de Los Muertos celebration, by having characters change costumes at a Halloween party, or by having aliens race across space. This structure, where content has already decoupled from context to support the creation of three thematic ways to teach a given concept, makes it an ideal candidate for harmonizing.

Professional Development Context and Teachers

The harmonized Scratch Encore materials presented in this work were created by teachers who participated in a weeklong, virtual summer professional development (PD) workshop. The PD consisted of 90-minute synchronous meetings each morning, during which teachers were introduced to the ideas and practices associated with harmonizing. This included being shown the various scaffolds designed to support the harmonizing process. At the end of the 90 minutes, they were given homework to be completed before the next day's session. This homework usually involved carrying out some aspect of the harmonizing process (e.g., ideate three potential harmonizing ideas for module 3 of Scratch Encore). By the end of the weeklong PD, teachers had harmonized two complete Scratch Encore modules. The teacher-created harmonizing examples shared in this manuscript were collected from teachers during this PD, which was led by this paper's authors and additional members of the larger research team.

The harmonizing Scratch Encore PD was open to any teacher who had previously taught Scratch Encore in their classroom. Twenty-four teachers participated in the harmonizing Scratch Encore PDs over the two summers it was offered (17 teachers in year one, seven teachers in year two). Seventeen of the teachers identified as female, six as male, and one did not specify. The participant pool was racially diverse, with 10 teachers self-identifying as White, four as Hispanic or Latina/o/x, four as Black or African American, three as Asian, one as Native American, and two not specifying. Teachers participated from across the United States, representing nine states, including California, Texas, Georgia, Illinois, Maryland, and Connecticut. The teachers who participated in this study had an average of 15.3 years of overall experience and an average of four years of teaching computer science. All recruitment and data collection activities were reviewed and approved by the IRB of the lead author's institution, and all participants signed consent forms to participate in the project.

THE HARMONIZING PROCESS

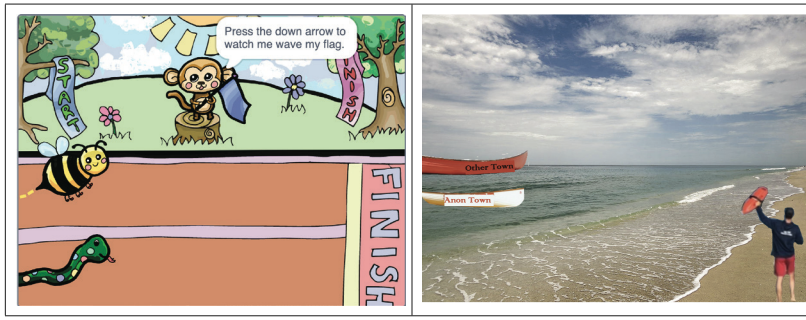
The main activity of the harmonizing process is to redesign an existing curriculum, instructional module, or activity so that it better resonates with the students who will be learning with it. This is accomplished by drawing on students' assets, including interests, prior knowledge, lived experiences, and their local community, as a means to make the context more famil-

iar and engaging. The goal is to keep the learning goals and pedagogical framework of the original activity fixed, but create a new contextualization of the materials. This way, the context of the lesson is customized to the students while the content remains the same. The three steps of the harmonizing process, shown in Figure 1, are ideation, implementation, and evaluation. For each step, we have developed a series of scaffolds and strategies to facilitate completing the process.

Before presenting the steps in detail, we begin this section by providing a concrete example of a harmonized lesson to provide a picture of what the endpoint of this process can look like. Figure 3 shows two Scratch projects that are used to teach students the concept of loops through animating sprites in Scratch. The original project on the left is taken directly from Scratch Encore and shows a race between a bee and a snake with a monkey waving a flag to start the race. On the right is a harmonized version of the project created by a teacher who teaches at a school in a beach community where the waterfront is a central part of life in the town. In the harmonized project, the bee and snake have been replaced with two lifeguard boats, and the monkey waving a flag has been replaced with a lifeguard waving a life preserver. While the original project is a race between two animals, the new, harmonized project is a race between two lifeboats moving toward the shore, recreating a popular lifeguard boat race held annually in the community. Additionally, the names on the sides of the boats were changed to be the local community and the next town up the shore, who regularly compete against each other in lifeguard games. Critically, both projects demonstrate the same technical content (animation via loops), and the underlying code for the two projects is identical: two sprites moving from left to right on the screen using a repeat loop, position changes, and multiple costumes, and a third sprite waving a flag/life preserver using a repeat loop and multiple costumes. The only things that have changed are the visual characteristics of the project, making it localized and more meaningful to students. When asked why she chose this topic, the teacher who created this project responded: “We are a beach community with a majority of our students participating in Jr. Lifeguard Training Camps during the summer. … The competitions that the Lifeguards have is also a big part of their summertime activity.” In this way, the teacher is able to connect the lesson’s content (e.g., animating sprites in Scratch) with a popular local activity that has meaning to the students in her class (e.g., lifeboat races at the beach).

Figure 3

An Example of a Harmonized Project



Note. Left panel: A Scratch project from Scratch Encore depicting a race between a bee and a snake. Right panel: A harmonized Scratch project that replaces the bee and snake with lifeboats from the local community, which situates the curricular module in the local community where the lesson is being taught.

Harmonizing Step 1: Ideation

The first step in the harmonizing process is coming up with a theme or context that will resonate with the students who will complete the instructional activity. For the teacher in the example above, the idea was to recreate a local lifeguard boat race as the context in which to introduce the concept of animation. As discussed above, research shows that compelling and familiar contexts are important for effective learning (NRC, 2018). As such, the central goal of the ideation step is to identify a context that will resonate with the interests and identities of the students who will complete the lesson and enable them to draw on their prior knowledge and lived experiences. The ideation step draws on the expertise of the teacher and their knowledge of their students, school, and community to create a customized and localized learning experience for their classroom. This is the key insight that led to the creation of the harmonizing process - teachers know their students better than anyone else, and thus, are best positioned to select topics, themes, and contexts that will resonate with them.

To facilitate the ideation step, we have developed three strategies, which differ based on where the ideas are being sourced from. The first ideation strategy sources themes from the teachers themselves. The second

seeks to draw ideas from the students and then has the teacher review and choose one that fits the instructional activity. The third uses generative AI as a thought partner to develop potential themes for the harmonized activity. A key objective across the three strategies is to select a theme that will resonate with students.

Ideation Strategy: Teacher-Sourced Ideation

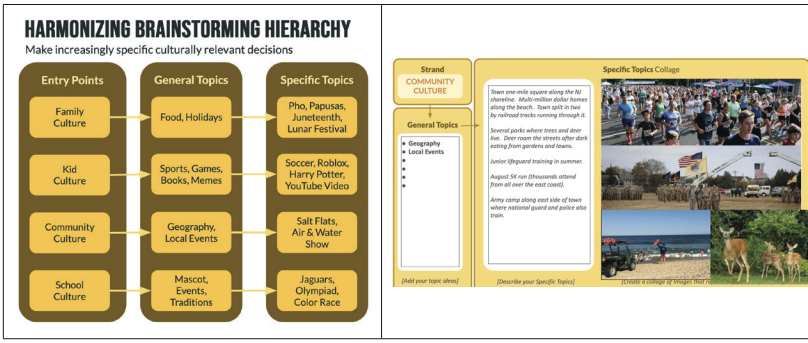
In teacher-sourced ideation, teachers brainstorm potential ideas for harmonizing. To facilitate this step, we developed a conceptual framework and scaffolds to structure teachers' brainstorming process. Through participatory design sessions with veteran teachers, we learned it is useful to provide a series of scoping prompts. Prompting teachers helps them identify general ideas, which they can systematically revise into more specific and ultimately, more implementable harmonizing ideas. Figure 4 shows the harmonizing brainstorming hierarchy framework we developed to facilitate this process. The highest level of this framework (e.g., entry points) suggests four potential sources of themes for a harmonized project: family culture, kid culture, community culture, and school culture. These four high-level groups are meant to serve as potential entry points in the ideation process and were identified after qualitatively coding teacher-generated ideas in participatory design workshops with the research team. The next level, "general topics," is where we provide a subset of themes that fall within that high-level category. For example, this would include moving from "community culture" down to "geography and local events," or from "kid culture" to "sports, games, books, memes." The goal here is to make more concrete but still relatively general suggestions for potential themes. The third and final level of the framework is "specific topics," which could serve as potential themes for a harmonized activity.

To accompany this framework, we developed a worksheet with a basic set of prompts to help teachers move from general to specific topics. During the PD, we had teachers ideate with this framework and create a collage of potential harmonizing ideas for each high-level entry point. Figure 4 includes the community culture collage created by a teacher who later developed the lifeboat race project presented in Figure 3. In describing her community culture collage, the teacher discusses the community's location (e.g., one-mile square along the state's shoreline, multimillion-dollar homes along the beach, town split in two by railroad tracks running through it.), important community landmarks and organizations (e.g., Army camp along

east side of town where national guard and police also train) and local community activities, including an annual community 5K run and the junior lifeguard training. This final idea, the junior lifeguard training program, was what she ultimately chose to use for harmonizing her Scratch project. In fact, the lifeguard in the picture from the collage was used in the harmonized project.

Figure 4

Ideation Strategy: Conceptual Framework and Scaffolds



Note. Left panel: The Harmonizing Brainstorming Hierarchy. Right panel: The Community Collage that led to the lifeboat race project presented above.

Ideation Strategy: Student-Sourced Ideation

The second ideation strategy crowdsources the idea generation process by having students generate potential ideas for themes to situate the instructional materials that will resonate with themselves and their peers. In this strategy, the teacher designs an activity for students to share information about themselves, their cultures, and their interests, then uses these ideas to inform the selection of a harmonizing theme. This strategy draws inspiration from the participatory design methodology, where end users, in this case, students, are included in the design process. This strategy is effective for incorporating the voices, values, and ideas of users into the co-designed artifacts (DiSalvo et al., 2017; Druin, 2002) and has a history of being used for the design of culturally responsive instructional materials (Coenraad et al., 2022). The strategy of having students share information about their

own interests and backgrounds is particularly valuable when teachers are unfamiliar with students or the larger community. Additionally, this strategy allows teachers to gain insights into current or popular student interests, such as trending memes or media, that can change rapidly.

The design activities teachers use to collect ideas from students can be brief and direct or creative and open-ended. For example, in our own harmonizing efforts, we used two distinct strategies to collect student ideas. The first strategy was to have students complete a survey about themselves, which was a straightforward way to learn about them. The survey included questions about their interests, like books and movies, hobbies, and activities they engage in with their families. The second strategy we used was asking students to create "about me" Scratch projects. In their projects, students shared their interests and their communities. As part of the PD workshops, we asked teachers to create about me projects as if they were students. Figure 5 shows three examples of about me projects that the teachers created, each of which sheds light on teachers and their interests, which provides insights that could serve as potential harmonizing themes if they were the students we were designing for.

Ideation Strategy: Generative AI as Thought Partner

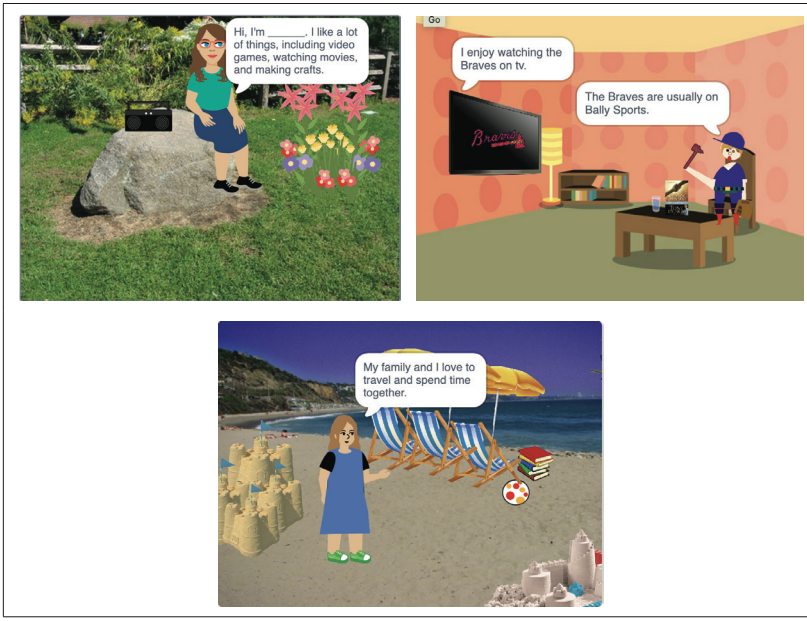
The third ideation strategy draws on the strengths of generative AI as a thought partner to ideate potential harmonizing themes. In this strategy, the teacher interacts with a generative AI tool to generate themes, and then the teacher reviews the AI-suggested themes and ultimately decides on a theme to incorporate into their harmonizing projects. This strategy builds on emerging scholarship on the utility of generative AI as a creative thought partner in educational contexts (Creely & Blannin, 2025). Our exploration of this strategy had teachers using existing off-the-shelf AI chatbots as thought partners (Tran et al., 2025). In the basic chatbot version, simple prompts to the generative AI tool yielded useful ideas for the teachers.

In introducing this strategy to teachers during the PD, one teacher produced the following prompt: "I need project ideas for a 5th grade level animation coding project with Scratch. Provide several topics that would be relevant to a class that is all girls, primarily low-income, from the Northwest side of the city. Our classroom is predominantly African- American, Norwegian-American, and Mexican-American. The projects need to have the following requirements: A single sprite that you can animate using a repeat loop and multiple costumes; A pair of sprites that you can both animate (using a repeat loop and multiple costumes) and move across the

stage (using a repeat loop); A great background." Figure 6 shows one of the five ideas offered by ChatGPT when given this prompt, which shows how such tools can be used for ideation and facilitate the task of coming up with a theme and then mapping it onto the technical constraints of the lesson.

Figure 5

Examples of About Me Projects that Share Creators' Interests



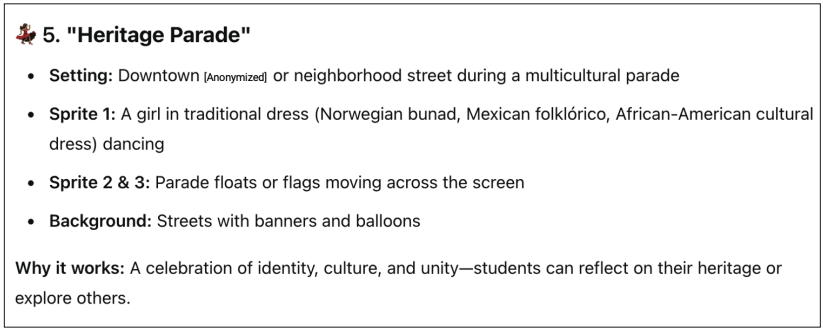
Harmonizing Step 2: Implementation

Once a teacher comes up with a theme for harmonizing a lesson or activity, the second step is to revise the existing materials to be situated within the new, harmonized context. While the details of exactly what it means to implement the harmonized lesson will vary based on the source lesson, the central goal remains the same: to keep the core learning goals and lesson structure fixed while changing the context. The first step of this process is to identify the learning goals and pedagogical structure of the lesson. The next step is to figure out how the new theme can be integrated into the lesson while preserving these existing key characteristics. Once the

strategy for operationalizing the theme has been decided, the teacher then goes through and creates the new instructional materials that will be used in the classroom.

Figure 6

ChatGPT's Suggestions for Harmonized Themes Based on Teacher-Authored Prompt



One productive strategy we have employed in our work is to work through the source materials and clearly state the focal learning goals of each activity and then identify how and where the context is defined. Practically, this means working through the various activities and worksheets students use and highlighting the places where the original context is articulated or referenced. Then, when going through and replacing the context with the new harmonized context, it was a matter of reviewing each of the highlighted parts of the source lesson and then revising them to fit the new, harmonized context. The left panel of Figure 7 shows two scaffolds we created for teachers to help with the implementation step of the harmonizing process. The right panel of Figure 7 shows part of a worksheet that highlights the learning goals of the lesson and how those concepts are presented to students. The left panel shows part of a worksheet that teachers complete to help plan out their implementation.

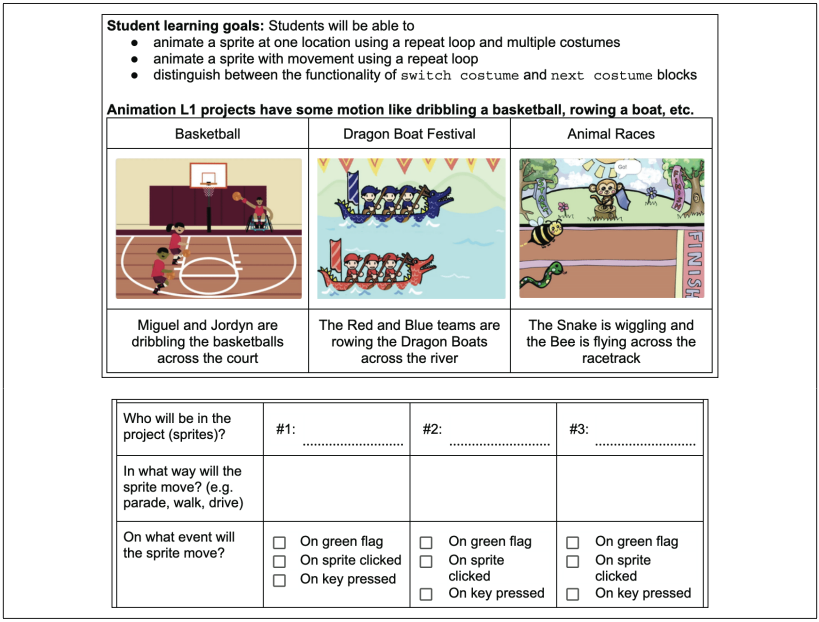
Harmonizing Step 3: Evaluation

The final step of the harmonizing process is to review the harmonized materials alongside the original materials to ensure the focal learning goals

are still being achieved without significantly changing the level of difficulty or complexity for students. This final step was added after we worked with teachers who harmonized instructional materials and found the harmonized materials significantly more complex than the original materials. In response to this, we developed a basic set of evaluation materials that let teachers quickly and easily compare their harmonized materials with the original materials. The goal of the evaluation materials is to assess the complexity of the harmonized materials, identify if the original learning goals are covered, and whether they introduce additional demands for students.

Figure 7

Two Teacher Scaffolds to Help with Implementation Step of the Harmonizing Process

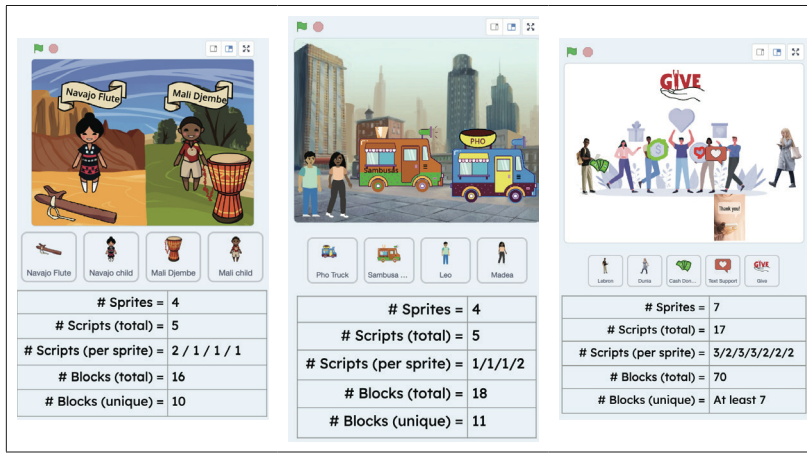


For Scratch Encore, we developed a basic scaffold we called the technical review worksheet to help teachers evaluate their harmonized Scratch projects in comparison to the source project. The goal was to help teachers see if and how their harmonized projects deviated from source projects in terms of concepts covered and overall complexity of the projects students would use. The technical review worksheet operationalized several sources

of complexity for Scratch projects, including the number of sprites, the number of scripts, the total number of blocks used, and the total number of types of blocks used. The right panel of Figure 8 shows the completed technical review worksheet for the one-way synchronization Scratch Encore project. The center and right panels of Figure 8 show two worksheets completed by teachers evaluating their harmonized projects for this module, one of which was comparable to the original (Figure 8, center panel) and a second which was significantly more complex (Figure 8, right panel).

Figure 8

Evaluating the Complexity of Three Scratch Projects



Note. Left panel: The Scratch Encore source project. Center panel: A harmonized project with similar levels of complexity. Right panel: A harmonized project with greater complexity.

DISCUSSION

Situating Harmonizing Relative to Other Instructional Models

This paper seeks to add the harmonizing process as yet another tool to the tool belt of teachers, teacher educators, and curriculum designers. While we think harmonizing is a unique process for customizing instructional materials, it does share some similarities with widely used instructional

models while also being distinct in meaningful ways. One meaningful distinction from many other instructional models is the fact that it is designed to start from an existing curriculum or set of instructional materials. Practically, this means it has less in common with frameworks such as ADDIE (Branch, 2009), Merrill's Principles of Instruction (Merrill, 2012), or Backwards Design/Understanding by Design (Wiggins & McTighe, 2005) and is instead closer to strategies seeking to modify existing materials. Two examples of similar instructional frameworks that can help situate the harmonizing process, both of which informed our conceptualization of it, are educational technology integration and cultural modeling.

The harmonizing process shares strategies with educational technology integration models such as the Substitution, Augmentation, Modification, and Redefinition (SAMR) model (Puentedura, 2013). In particular, the process of analyzing an existing set of instructional materials and identifying ways to systematically modify it without changing the underlying instructional goals is very similar to the core activity of the harmonizing process. Another parallel to harmonizing can be seen in Lee's (2001) Cultural Modeling framework. The central goal of the Cultural Modeling framework is to attend to prior knowledge and cultural models as ways of knowing of students to support engagement and motivation, as well as social and civic empowerment. Key steps in the Cultural Modeling framework also mirror the harmonizing process, including the initial step of analyzing the source academic domain to identify core learning goals and connecting them to learners' lives and the subsequent step of considering prior knowledge and cultural resources that can be drawn upon to situate the learning activity (Lee, 2003).

Implications for Teachers

The harmonizing process was designed for educators to customize and localize their instruction. This has direct implications for their lesson planning as it requires them to carry out the localization and customization efforts. In our work on harmonizing, teachers are introduced to the process and its steps as part of a summer professional development. This provides a natural space for teachers to learn about the motivation behind the process and the various steps associated with carrying it out. At the end of the five-day professional development session, teachers have two harmonized lessons that are ready for their classrooms. For teachers who are being introduced to harmonizing in a less formal way, there are still implications,

both practical and theoretical, for their teaching. The practical implications are the strategies and resources directly associated with the harmonizing process, in whatever form they choose to enact it. More generally, being introduced to the basic idea and goals of harmonizing can inform their pedagogy in less direct ways. For example, being aware of the importance of situating instruction in ways that resonate with students' lived experiences may lead to teachers paying closer attention to student interest or keeping an eye out for opportunities to bring students' lives or aspects of the school community into the classroom. This may result in minor shifts in instruction or lesson design that invite personalization and give opportunities for learners to share a bit more of themselves and their lives as part of classroom activities. In this way, the idea of harmonizing doesn't lead to teachers creating harmonized materials directly but instead designing opportunities that lead to the same end—students situating the content they are learning relative to their interests, identities, and lived experiences.

Implications for Teacher Educators

Building on the implications for educators, the harmonizing process also has implications for teacher educators. For preservice teacher education courses, introducing future teachers to the harmonizing process can serve as a demonstration of what it looks like to begin to customize instructional materials for their classrooms. Additionally, as future teachers will have learned about the importance of aligning instruction with students' prior knowledge and interests, this harmonizing process can serve as a concrete example of how they might go about accomplishing these goals. Likewise, it can provide an initial, scaffolded experience of trying to modify a lesson they might teach in their future classroom.

For continuing education contexts, where teacher educators are working with in-service teacher educators, the harmonizing process can be an effective way to help teachers reflect on and modify their current teaching materials. A teacher who has already taught a lesson a few times will know the strengths and drawbacks of the lesson and will also know the characteristics of their students, school, and larger community. This is an ideal audience for the harmonizing process, as the teacher will already have the prior instructional experiences as well as a clear, concrete context to design for. This situation is closest to our own work with experienced Scratch Encore teachers.

Finally, at a more meta level, it is possible to envision a teacher educator using the harmonizing process to modify their own instructional

approach in their teacher education courses. For example, this might mean harmonizing class activities to align with specific features or challenges they know their students will face when they enter the classroom. Likewise, it could mean redesigning activities in their teacher education courses, so teachers have more flexibility to integrate aspects of their future teaching placements or future teaching responsibilities into the work. This may be particularly beneficial for classes where there is little uniformity in the future teaching responsibilities of the students (e.g., differing disciplines, differing age groups, differing school districts/geographic regions).

Implications for Curriculum Designers

The harmonizing process has several implications for curriculum designers. First and foremost, among them is thinking about ways to design instructional materials that lend themselves to being harmonized. By this, we mean creating curricula where the content and learning objectives are easily decoupled from the theme in which the content is situated. The Scratch Encore project serves as a nice example of what this might look like due to its multi-theme strand design. For curricula where it is not possible, or not easy to carry out a harmonizing process due to the tight coupling of content and context, this work can help highlight the importance of acknowledging the localized expertise that teachers have and considering how to structure instructional materials such that teachers can draw on that expertise. Based on this recognition, designers can start to build into their curricula opportunities that invite them to draw on that expertise in their teaching. For example, it is easy to envision a lesson or activity that explicitly builds in some sort of harmonizing features. By this, we mean it asks teachers to choose a theme that will resonate with their students or invites them to ask their students for a theme and then uses that as part of the instructional approach. This strategy may be particularly effective for disciplines where the theme or context is not necessarily tightly coupled to the content, as is the case with computer science or mathematics.

Drawing on Teacher Expertise Without Overwhelming Teachers

A core insight that motivates this work is that teachers are experts on their own students, schools, and communities, and thus, are best suited to create harmonized materials for their students. However, teachers rarely

have the bandwidth to fully use this localized expertise to inform the design of the materials they teach due to limited time, resources, and generally being overworked. In our early efforts to support the harmonizing process, we saw this limitation firsthand and documented several challenges teachers encountered while trying to harmonize, including the process taking too long, difficulty coming up with ideas that would resonate with students, and struggling to retain the original learning goals during the harmonizing process (Tran et al., 2024). In response to this feedback, we conducted a series of iterative design cycles to develop scaffolds and strategies to help teachers along each of these dimensions. In bringing these innovations back to teachers, we found they helped teachers complete the harmonizing process while also allowing them to draw on their localized expertise, creating projects situated in their communities or their students' cultures.

CONCLUSION

Students come into the classroom with a wealth of prior experiences and knowledge that can be utilized as part of instruction to make content more accessible and engaging. No one is in a better position to draw on these resources than the teacher standing at the front of the classroom. However, creating customized and localized instructional materials can be challenging and time-consuming. Towards these challenges, the driving research question for this work asked: How can we support teachers in creating customized, localized, culturally responsive instructional materials that draw on the prior knowledge and cultural resources present in their classrooms? The answer this work presents is the harmonizing process, a three-step process that walks teachers through developing ideas for themes they can use to situate instruction to resonate with students, integrating those ideas into instructional materials, and ensuring the resulting materials meet the same learning objectives as the source materials.

This work introduces the harmonizing process to help teachers modify existing instructional materials to better draw on the prior knowledge and experiences of students in their classrooms. The primary contribution of this work is providing a framework, along with accompanying language and strategies, to support teachers in carrying out the important work of tailoring classroom instructional materials to draw on the prior knowledge and cultural resources present in their classrooms. Collectively, this work seeks to provide guidance and support to educators who wish to create customized, localized, and personalized learning experiences for their students.

The result of this process is teacher-customized instructional materials that reflect the unique contexts of their students and classrooms. In doing so, this work empowers teachers to create instructional materials to help their students reach their full potential.

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